NLP Study Notes

Natural Language Processing · Architecture · Preprocessing

1 · Model Architecture

Encoder–Decoder / Seq2Seq

Context Understanding

Models that understand the context of speech by encoding an input sequence into a fixed-length context vector, then decoding it into an output sequence. The canonical example is machine translation (e.g., English → French).

■ **Key Paper:** *Attention Is All You Need* (Vaswani et al., 2017) — introduced the Transformer architecture based entirely on self-attention, replacing recurrence and convolutions.

■ **More tokens ≠ bad.** Generating more tokens is *beneficial* for reasoning — extended generation lets the model work through intermediate steps before producing a final answer.

2 · NLP Preprocessing Pipeline

Preprocessing converts raw text into a form suitable for modelling. The mandatory step is **Tokenization** (text → tokens). All subsequent steps are task-dependent.

Typical Pipeline Order

Tokenization	Stop Words	Stemming / Lemmatization	Regex	POS	NER	Spell Correction

Mandatory: Tokenization · Optional (task-dependent): everything after

Phases

Phase	Steps Included
Preprocessing	Tokenization · Stemming or Lemmatization · Regex

Annotation	POS (Part-of-Speech) · NER (Named Entity Recognition)
Layer Correction	Spell Correction

1. Stop Words

Reduce Noise

High-frequency, low-semantic-value words such as *"the"*, *"is"*, and *"and"* that are removed to reduce noise and computation time.

■ Works well for Sentiment Analysis, but **not** for Chatbots — removal changes sentence context and breaks meaning.

2. Stemming

Root Reduction

Reduces words to their base/root form by chopping off inflections such as *-ing*, *-ed*, or *-s*. Fast but crude — the result may not be a real word.

✔ Common use case: Sentiment Analysis.

3. Lemmatization

Dictionary Form

Reduces a word to its **lemma** — its valid dictionary entry — using vocabulary and morphological analysis. Unlike stemming, the output is always a meaningful word. More accurate, but slower.

✔ Example: *running* → *run* (lemma) vs *runn* (stem).

4. Regular Expressions (Regex)

Text Cleaning

Used for pattern-based text cleaning: removing URLs, punctuation, special characters, whitespace normalisation, etc.

5. Part of Speech (POS)

Syntax Tags

Assigns grammatical categories (noun, verb, adjective, …) that define the role each word plays in a sentence based on its syntax and meaning. Feeds into parsing and NER.

6. Named Entity Recognition (NER)

Entity Extraction

Identifies and classifies specific information — *entities* — within text into predefined categories:

Tag	Category	Examples
`PER`	Person	*Albert Einstein, Sherlock Holmes*
`ORG`	Organisation	*Google, United Nations*
`LOC/GPE`	Location	*Paris, Brazil*
`DATE/TIME`	Date & Time	*5 May 2025, 3:00 PM*
`NUM`	Numerical	*10 kg, 50%, $100*

7. Spell Correction

Error Fixing

Automated identification and correction of typographical errors to improve accuracy for both humans and downstream models.

3 · Optical Character Recognition (OCR)

OCR converts images containing text — scanned paper documents, photos of signs, image-only PDFs — into machine-readable, editable text data. It is typically applied *before* the NLP preprocessing pipeline when the source is non-digital.

■ Image / Scan → ■ OCR Engine → ■ Plain Text → ■ NLP Pipeline

4 · Word Embeddings & Word2Vec

Word Embedding maps words (or phrases) to dense numerical vectors in a continuous vector space, capturing semantic relationships. Words with similar meanings are closer in this space.

Method	Description
`Word2Vec`	Shallow two-layer neural network that learns word associations from a large corpus. Produces vectors where arithmetic on embeddings reflects semantic relations (*king* – *man* + *woman* ≈ *queen*).
`Skip-Gram`	A Word2Vec variant that predicts surrounding context words given a centre word. Works well with smaller datasets and learns better representations for rare words.

■ Word2Vec → Skip-Gram is the standard starting point for understanding how modern NLP models represent meaning before attention mechanisms.